



CAPITAL HEALTH

Hospitals

A Data Driven Approach to Better Care

Vanessa Negrete-Vega, Neha Malviya, Nishtha Patel, Manavi Yadav



capitahealth

AGENDA

1

OVERVIEW

2

ANALYSES

3

FINDING AND INSIGHTS

4

RECOMMENDATIONS

5

USE OF AI

6

REFERENCES

Important Variables

Account No.

The primary key that makes everything else work

Diag10

Tells why the patient was there

Admit Dt/Disch Dt

Primary Variable for time analysis across different branches

About the Data

- **Four files** — HPW Adult, HPW Peds, RMC, Deborah
- **Time period** — March 2021 to February 2022
- **Each row = one unique patient visit**
- **Capacity file** adds arrival and departure times
- **Deborah** — outpatient only, no overnight stays

Data Cleaning



Missing Values



Summary Statistics



Code Translation



Visual Validation

Data Cleaning



Formatting



Column Splitting

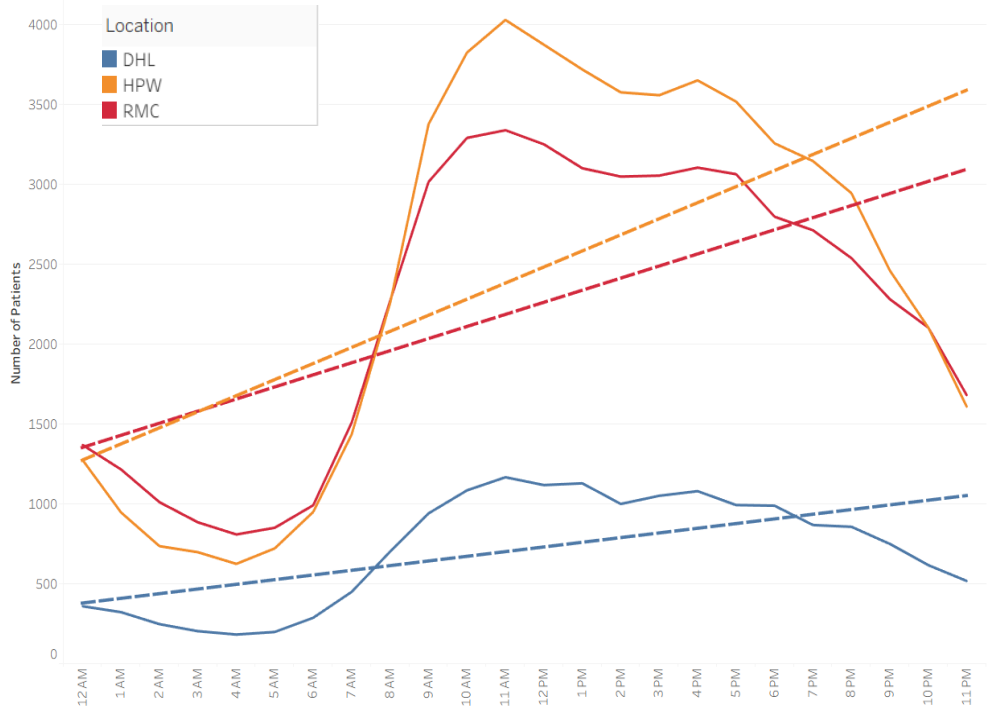


Chronological Validation



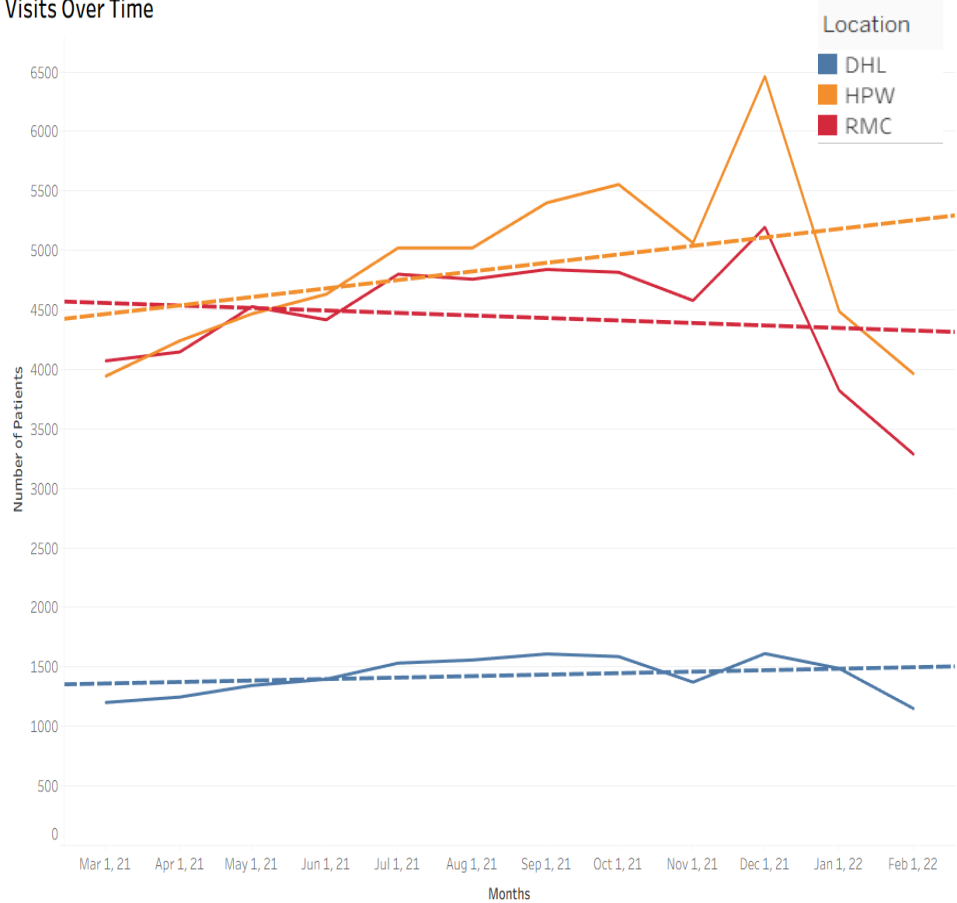
Duplicate Check

Visits by Hour



Campus	Peak hour	Peak Volume	Lowest Hour	Lowest Volume	Trend
RMC	11 AM	~3,300	4–5 AM	~750	Upward
HPW	10–11 AM	~4,050	4–5 AM	~625	Upward
DHL	11AM – 12 PM	~1,175	4:00 AM	~125	Slight upward

Visits Over Time



Campus	Peak Month	Peak Volume	Lowest Month	Lowest Volume	Trend
RMC	Dec-21	~5,200	Feb-22	~3,300	Declining
HPW	Dec-21	~6,500	Mar-21	~3,950	Upward
DHL	Jun–Aug 2021	~1,600	Mar 2021 / Feb 2022	~1,175	Flat / stable

ARIMA MODEL

1

DHL → ARIMA(0,1,3): driven purely by past random shocks (noise only)

2

HPW → ARIMA(4,1,3): driven by both past admission patterns (signal) and random shocks (noise)

3

RMC → ARIMA(2,1,2): driven by both signal and noise, with strong day-to-day persistence

```
Series: DHL
ARIMA(0,1,3)

Coefficients:
      ma1      ma2      ma3
    -0.786  -0.1793  0.1114
s.e.   0.052   0.0657  0.0530
```

```
sigma^2 = 76.62: log likelihood = -1305.33
AIC=2618.67 AICc=2618.78 BIC=2634.26
> auto.arima.HPW <- auto.arima(HPW)
> auto.arima.HPW
```

```
Series: HPW
ARIMA(4,1,3)
```

```
Coefficients:
      ar1      ar2      ar3      ar4      ma1      ma2      ma3
    -0.0091  0.1224  -0.5138  -0.3531  -0.5910  -0.2669  0.5450
s.e.   0.1700  0.1687  0.0952  0.0636  0.1727  0.2745  0.1358
```

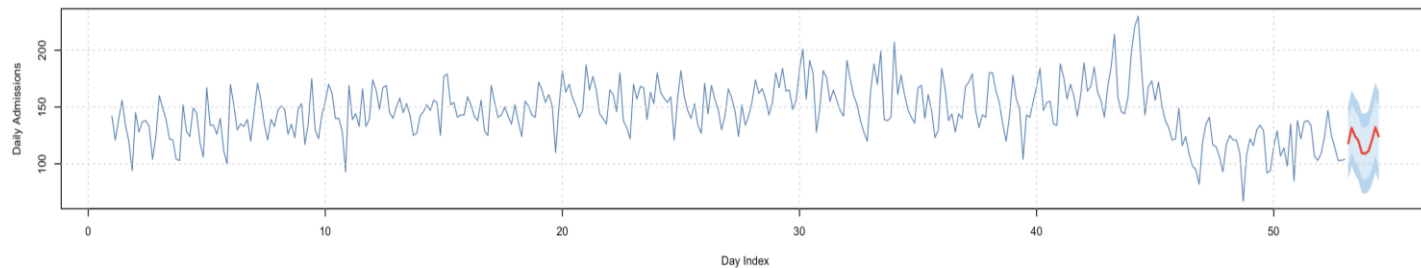
```
sigma^2 = 462.8: log likelihood = -1630.79
AIC=3277.57 AICc=3277.98 BIC=3308.75
> auto.arima.RMC <- auto.arima(RMC)
> auto.arima.RMC
```

```
Series: RMC
ARIMA(2,1,2)
```

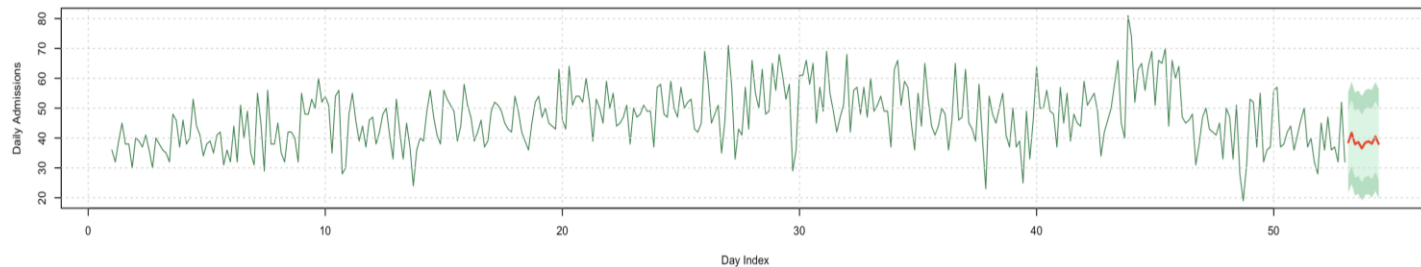
```
Coefficients:
      ar1      ar2      ma1      ma2
    0.8969  -0.4535  -1.4851  0.6310
s.e.   0.0811  0.0666  0.0714  0.0722
```

```
sigma^2 = 336.6: log likelihood = -1574.18
AIC=3158.35 AICc=3158.52 BIC=3177.84
```

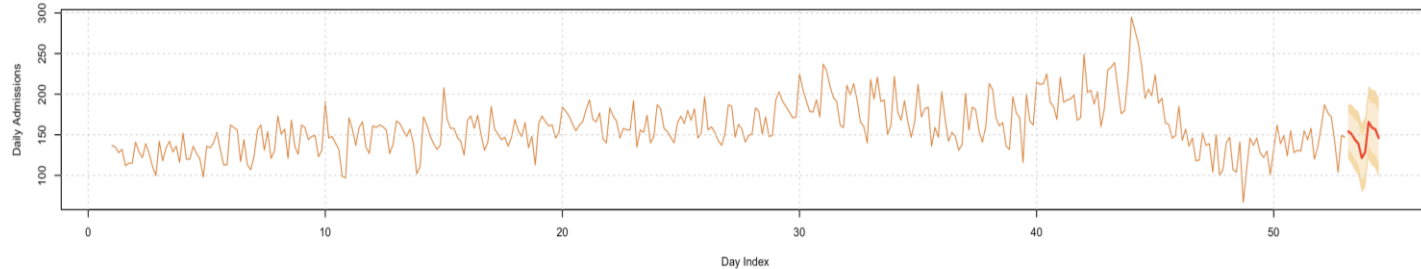
RMC — 10-Day Forecast



DHL — 10-Day Forecast



HPW — 10-Day Forecast



Sentimental Analysis Review

- 1 — RMC (Negative) ★★☆☆☆** "Waited over 4 hours in the ER and not a single nurse came to update me. When I finally got seen, the care was fine — but the wait was unacceptable. I left feeling like a number, not a patient."
— Patient review, Capital Health RMC, 2021
- 2 — RMC (Negative) ★★☆☆☆** "I received a \$1,200 bill with no breakdown and no explanation. When I called billing, no one could tell me what I was charged for. The doctor was great but the experience around it was a nightmare."
— Patient review, Capital Health RMC, 2022
- 3 — Hopewell (Mixed) ★★☆☆☆** "The facility was clean and the doctors were knowledgeable. The ER wait was long but manageable. Billing had a small error they fixed quickly. Overall decent, but room to improve."
— Patient review, Capital Health Hopewell, 2021
- 4 — Hopewell (Positive) ★★★★★** "Had my baby here and the maternity team was wonderful — attentive, kind, and professional throughout. The postpartum nurses checked on me constantly. Would absolutely come back."
— Patient review, Capital Health Hopewell, 2022
- 5 — Deborah (Positive) ★★★★★** "The cardiac team at Deborah is in a league of their own. Every single nurse knew my chart. The doctors explained everything clearly. I never once felt rushed or ignored."
— Patient review, Deborah Heart & Lung Center, 2021
- 6 — Deborah (Positive) ★★★★★** "After two bad experiences at other hospitals, Deborah was a completely different world. Immaculate facility, zero wait, and a care team that genuinely felt invested in my recovery."
— Patient review, Deborah Heart & Lung Center, 2022

**BIGGEST WEAKNESS**

ER wait times hit -0.55 at RMC

Worst-scoring dimension across all campuses

UNIVERSAL PAIN POINT

Billing is negative at every campus

Only theme where even Deborah scores below zero

DEBORAH'S STRENGTH

Cardiac care scores $+0.85$

Far outperforms RMC & HPW on clinical dimensions

HIDDEN RISK

Employee culture negative at all 3

Staff dissatisfaction drives future patient experience decline

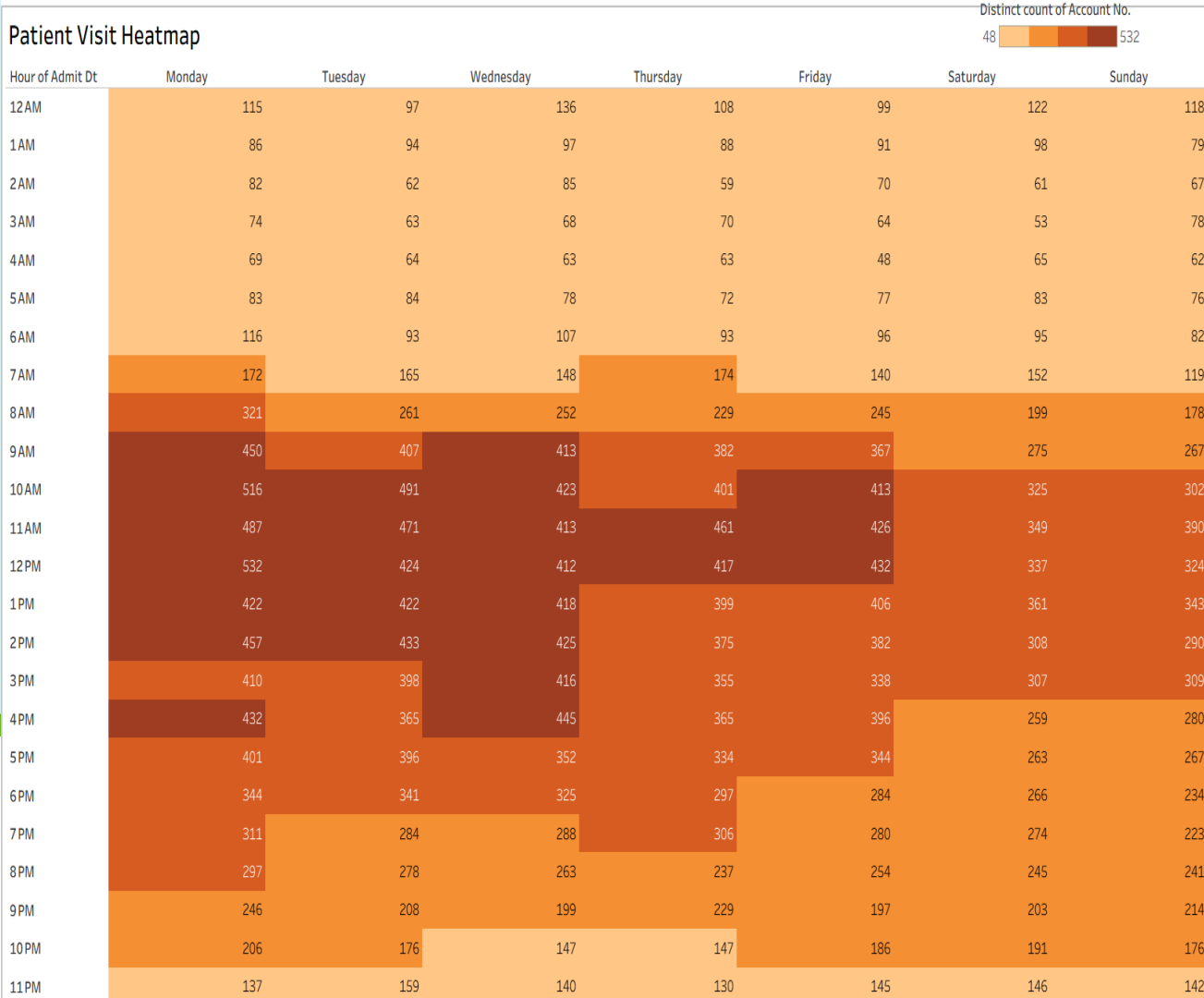


Patient Visit Heatmap



RMC

RMC sustains peak-level admissions from 9 AM through 6 PM daily — the heaviest and most prolonged demand of all three campuses. This sustained overload is the root cause of the -0.30 composite sentiment score, not clinical failure.



HPW Adult

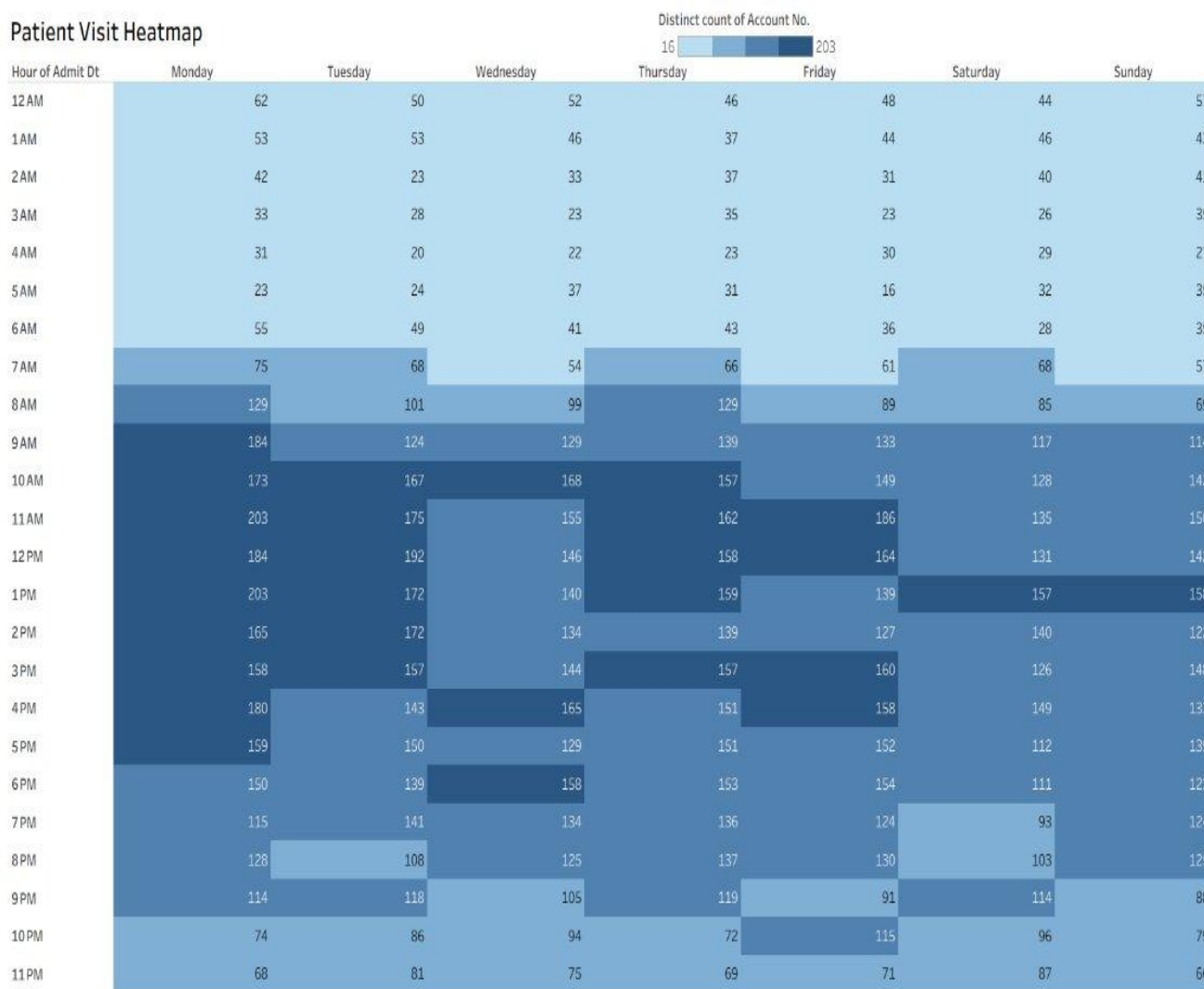
Hopewell Adult follows a similar weekday pattern but the volume is concentrated more in the morning — roughly 9 AM to 1 PM. Afternoons are lighter.

Patient Visit Heatmap



HPW Peds

Pediatric visits peak later — 11 AM through 5 PM — and stay elevated on Mondays and Thursdays. Evening staffing is critical for this unit, unlike adult admissions.

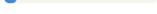


DHL

Deborah's volume is lower and spread evenly across the day with no extreme spikes. This steady flow allows staff to provide consistent, attentive care — directly reflected in its nursing score of +0.55 and clinical quality of +0.75.

Sentimental Analysis

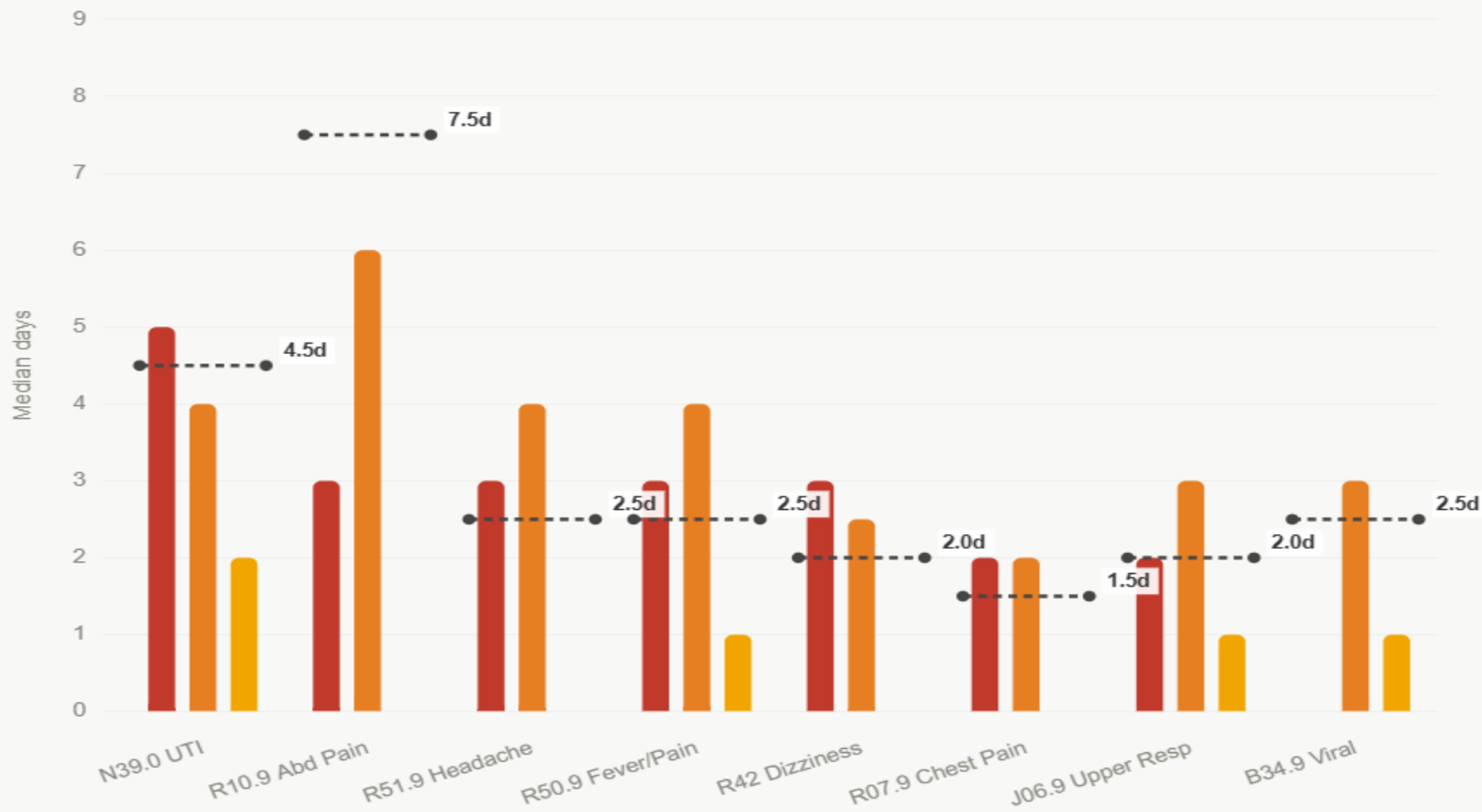
✓ Accepted ✗ Not listed ~ Select plans only

Insurance plan	NJ enrollment	Cap Health RMC	Cap Health HPW	Deborah H&L
Medicare Traditional + Medicare Advantage	 1.7M	✓	✓	✓
Horizon BCBS NJ PPO, HMO, Horizon NJ Health	 1.6M	✓	✓	✓
Medicaid / NJ FamilyCare State Medicaid managed care	 1.5M	✓	✓	✓
UnitedHealthcare (Medicaid) coverage gap UHC Community & State, NJ	 345K	✗	✗	✓
Aetna Commercial + Medicare Advantage	 291K	✓	✓	✓
UnitedHealthcare (Commercial) coverage gap Choice Plus, Choice, Select	 215K	✗	✗	✓
Cigna Open Access Plus, LocalPlus	 198K	✓	✓	✓
Amerigroup NJ coverage gap Medicaid managed care	 182K	✗	✗	✗
Oxford Health Plans (UHC subsidiary)	 ~130K	✗	✗	✓

Coverage gap: ~872,000 NJ residents affected

Capital Health (RMC + HPW) does not accept UnitedHealthcare, Oxford, or Amerigroup — plans held by an estimated 872,000 New Jerseyans. Deborah accepts UHC and Oxford, giving it a significant access advantage. This gap is a likely structural driver of lower patient satisfaction scores at Capital Health facilities in 2021–2022.

RMC HPW Adult HPW Peds National benchmark



Recommendations



Expand RMC Staffing Hours — Increase staff coverage between 9 AM–6 PM to address sustained peak demand and improve patient wait time satisfaction scores.



Overhaul Billing Transparency — Implement itemized billing statements and dedicated billing support staff to address the system-wide negative billing sentiment across all three campuses.



Extend HPW Pediatric Evening Coverage — Schedule additional nursing staff through 5–6 PM on Mondays and Thursdays when pediatric volume remains elevated.



Invest in Employee Culture — Launch staff satisfaction initiatives across all campuses, as negative employee sentiment poses a long-term risk to patient experience quality.



Negotiate New Insurance Contracts — Pursue contracts with UnitedHealthcare, Oxford, and Amerigroup to close the coverage gap affecting an estimated 872,000 NJ residents.



Redistribute Patient Load Using ARIMA Forecasts — Use the predictive admission models built for RMC, HPW, and DHL to proactively schedule resources and redirect non-urgent cases during peak periods

How we used AI

Data Cleaning & Presentation

1

Used AI to clean, organize, and structure large datasets for analysis.

Data Analysis Support

3

Used AI to help understand patterns, trends, and apply the right analysis methods.

Coding & Tools Assistance

2

Used AI to improve and write code in R, Excel, and other tools used in the project

Insights & Presentation

4

Used AI to create clear visuals and improve explanations for better storytelling.

References

- Cooperman Barnabas Medical Center. (2022). *Patient testimonials*. RWJBarnabas Health <https://www.rwjbh.org/cooperman-barnabas-medical-center>
- Deborah Heart and Lung Center. (2022). *Patient experience & testimonials*. <https://www.deborah.org>
- U.S. News & World Report. (2021–2022). *Best hospitals: Cooperman Barnabas Medical Center*. <https://health.usnews.com/best-hospitals>
- U.S. News & World Report. (2021–2022). *High performing hospitals for maternity care*. <https://health.usnews.com/best-hospitals/maternity>
- The Joint Commission. (2022). *Comprehensive stroke center certification: Robert Wood Johnson University Hospital*. <https://www.jointcommission.org>
- American Heart Association. (2022). *Get with the guidelines — stroke gold plus award*. <https://www.heart.org/en/professional/quality-improvement/get-with-the-guidelines>
- Indeed. (2021–2022). *Employee reviews: RWJBarnabas Health campuses*. <https://www.indeed.com>
- Glassdoor. (2021–2022). *Employee reviews: RWJBarnabas Health*. <https://www.glassdoor.com>
- Agency for Healthcare Research and Quality. (2022). *HCUP National Inpatient Sample (NIS)*. Healthcare Cost and Utilization Project. <https://www.hcup-us.ahrq.gov/nisoverview.jsp>
- Centers for Medicare & Medicaid Services. (2022). *Hospital compare: Patient survey results*. U.S. Department of Health & Human Services. <https://www.medicare.gov/care-compare>
- Tamma, P. D., Avdic, E., Li, D. X., Dzintars, K., & Cosgrove, S. E. (2017). Association of adverse events with antibiotic use in hospitalized patients. *JAMA Internal Medicine*, 177(9), 1308–1315. <https://doi.org/10.1001/jamainternmed.2017.1938>
- Healthgrades. (2021–2022). *Patient reviews: Robert Wood Johnson University Hospital*. <https://www.healthgrades.com>
- RateMDs. (2021–2022). *Patient reviews: Robert Wood Johnson University Hospital & Deborah Heart and Lung Center*. <https://www.ratemds.com>
- HospitalStats.org. (2021). *Patient reviews: Cooperman Barnabas Medical Center Hopewell*. <https://www.hospitalstats.org>



Thank You

Question 1: RMC has the most patients and the longest average stay. What does that tell us about RMC's role in the Capital Health system?

Yes, RMC clearly is the busiest and most demanding facility in the system. In RMC patients stay longer which means they are coming in with more serious conditions that take more time to treat. Looking at the types of diagnoses at RMC, things like heart and circulatory problems show up frequently, and they clearly need more treatment time. So RMC is not just busy in terms of number of patients, it is busy with harder cases. Through which we can interpret that RMC needs more staff, more resources, and more specialist coverage compared to the other three facilities.

Question 2: Deborah is part of the Capital Health system, so patients with those plans can still access care within our network. Why is this still a concern?

That's true, and it's an important clarification, however the concern is really about the volume, we're talking about an estimated 872,000 New Jersey residents whose plans aren't accepted at RMC or HPW specifically. Even within the same system, if patients are being funneled to one campus because of insurance limitations, that creates capacity and access imbalances. And it's also worth noting that Deborah specializes in heart and lung care, so a patient with UnitedHealthcare who needs something like an abdominal workup or a UTI treated isn't really being served by having access only to Deborah. They need RMC or HPW, and right now their insurance is a barrier to that.

Question 3: HPW Adult ED and Hopewell Pediatrics are under the same Hopewell branch. Can we manage them together with the same budget and staffing plan?

No, that would cause more problems. Even though they fall under the same branch name, these two clinics are for very different purpose of treatment. HPW Adult ED sees a large number of patients with serious, long-term conditions that require significant time and resources. Whereas, Hopewell Peds see children with shorter, more basic diagnosis causes. If we try to apply one staffing plan to both, we will either have too many staff at the pediatric clinic or not enough at the adult ED. It's the same for budgeting, combining their finances could increase the problem as the adult ED needs more money and resources. The right approach is to run each clinic based on what it actually needs.
